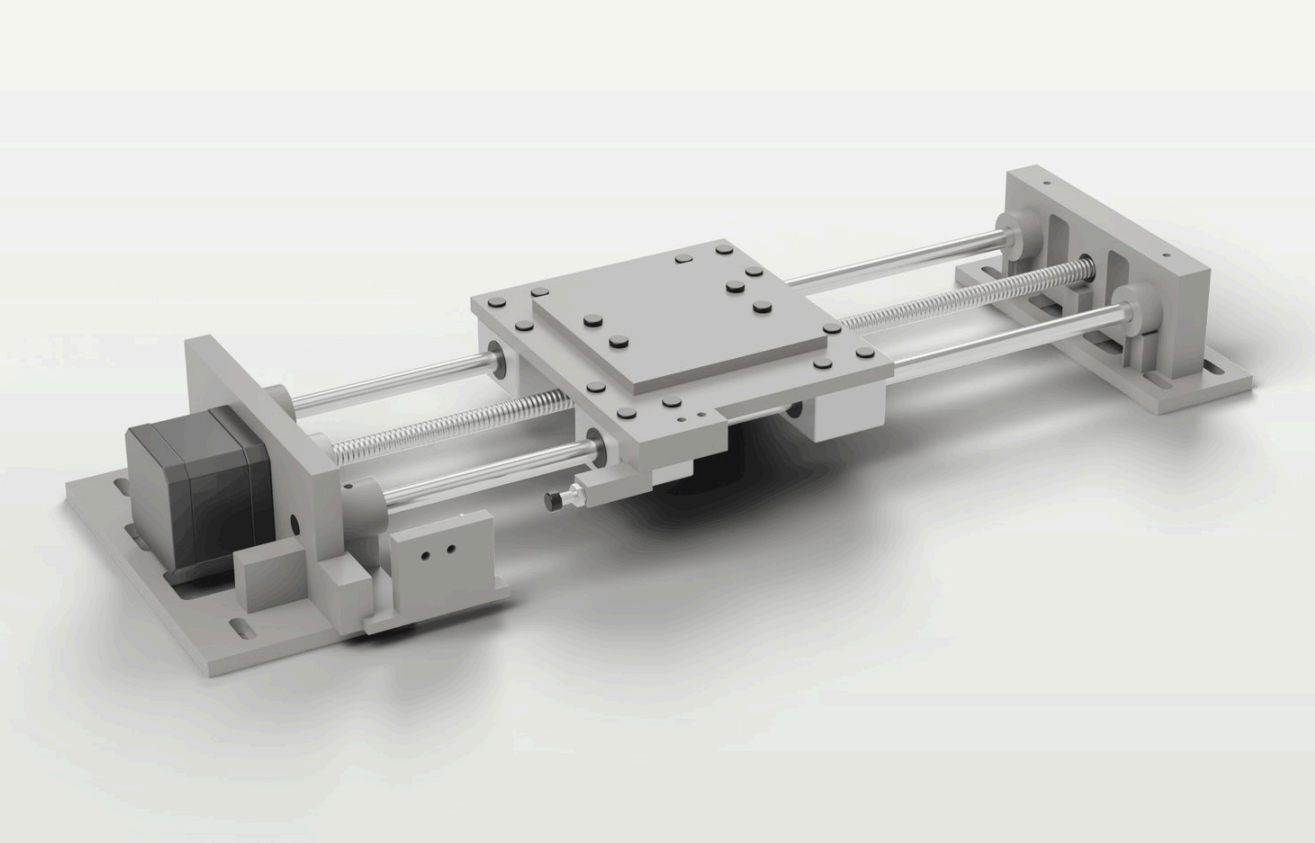
MACRORAIL

Build guide

A motorised focus-stacking rail: parts, printing, assembly, electronics, software and first use.



For the rail as published at github.com/Sam-de-Ham/MacroRail, November 2026. The optional scan module has its own guide.

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1 About the rail

MacroRail moves the subject along the lens axis in steps of a few micrometres while a fixed camera takes one photograph per step. Combining the photographs (focus stacking) gives one image that is sharp throughout. The rail is meant for macro lenses and microscope objectives, where the depth of field is a few tens of micrometres.

A Raspberry Pi runs a web app that is used from a phone or a laptop on the same network. The Pi talks to an Arduino Nano over USB; the Nano drives a NEMA 17 stepper through a TMC2209 driver and watches a home switch. The camera is connected to the Pi by USB and controlled with gphoto2.

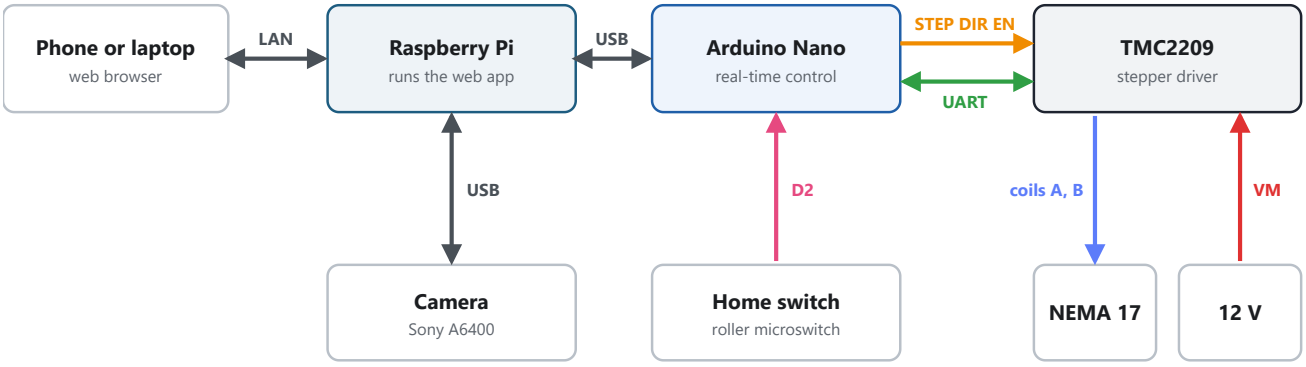


Figure 1. How the parts are connected. The browser never talks to the hardware directly: everything goes through the Pi.

Main figures of the rail as built.

Item	Value
Travel	140 mm usable (soft limit `travel_mm`); about 150 mm mechanical
Resolution	2.5 µm per microstep (Tr8×8 screw, 200 steps/rev, 1/16 microstepping)
Lead-screw backlash	14 µm ± 7 µm measured; avoided by always arriving in the stack direction
Speeds (defaults)	travel 2 mm/s, during a stack 1 mm/s, acceleration 10 mm/s^2
Motor current (defaults)	600 mA running, 200 mA holding, set over UART
Supplies	12 V for the motor; 5 V for the Pi (2.5 A); the Nano is powered by the Pi's USB
Footprint	about 400 × 140 mm, on a heavy base

1.1 How to use this guide

Work through the sections in order. Sections 2 to 4 cover the mechanics, 5 and 6 the electronics, 7 and 8 the software, and 9 to 11 the first power-up, the camera and the first stack. Section 12 lists common faults. Steps that matter for safety or for the electronics are marked as cautions.

> **Before you start**
> Never connect or disconnect the motor while the 12 V supply is on: it can destroy the TMC2209. Disconnect the power while assembling or adjusting, and keep fingers clear of the coupler and the lead screw whenever the motor is powered.

A simulator lets you try the app without any hardware: on a computer with uv installed, run `uv run macrorail --sim` in the repository and open http://localhost:8000.

2 What you need

2.1 Bought parts

Prices are approximate (euro, 2026, from general online shops) and only meant as a guide. Much of the motion hardware can come from an old 3D printer: the prototype used the rods, bearing blocks, lead screw and motor of an Anet A8.

Part	Qty	Specification	≈ €
Mechanics			
Smooth rod	2	Ø8 mm, 380 mm long	12
Linear bearing block	4	SC8UU (8 mm, M4 mounting holes)	12
Lead screw with nut	1	Tr8×8 (8 mm lead, 2 mm pitch, 4 starts), about 345 mm; flanged T8 brass nut (Ø22 flange) with its four M3 screws	10
Flexible coupler	1	5 mm to 8 mm, helical, about Ø19 mm	3
Stepper motor	1	NEMA 17, 1.8°, 4 wires, 5 mm shaft (prototype: 42SHDC3025-24B)	12
Roller-lever microswitch	1	3 terminals (C, NO, NC), e.g. KW12 type	1
Fasteners		M3, M4 and M5 screws, nuts and washers; see 2.3	8
PLA filament		about 350 g	7
Base	1	flat, heavy board or plate, at least 420 × 160 mm	10
Electronics			
Arduino Nano	1	ATmega328P; a CH340 clone is fine; with a USB data cable (Mini-B on most)	6
Stepper driver	1	TMC2209 module (StepStick format, BTT or FYSETC style) with the PDN_UART pin brought out; sense resistor 0.11 Ω	6
Motor supply	1	12 V DC, 2 A, with leads or a screw-terminal adapter	9
Capacitor	1	100 µF electrolytic, 25 V or more	0.2
Resistors	1 + 1	1 kΩ and 10 kΩ, ¼ W	0.2
Breadboard and wires	1	half-size breadboard (30 columns); male–male jumper wires in several colours; wire for the switch	6
Computer			
Raspberry Pi	1	3B+ or later (tested on a 3B+); microSD card, 16 GB or more; official 5.1 V / 2.5 A supply	55
Total, without camera and optics			**about 160**

Also needed, not included above: a camera that gphoto2 can trigger remotely (tested with a Sony A6400) with its USB cable, a lens or objective, a tripod or stand that holds the camera at the rail's height, and preferably a mains adapter (dummy battery) for the camera. A Pi under-supplied by a weak adapter drops USB devices; use a proper 2.5 A supply.

2.2 Printed parts

The six STL files are in `hardware/rail/stl/`. They keep their assembly coordinates, so use *drop to build plate* and *arrange* in the slicer.

File	Qty	On the bed	Infill
`01_motor_end_block`	1	broad mounting foot	40–60 %
`02_far_end_block`	1	broad mounting foot	40–60 %
`03_carriage`	1	largest flat face	40–60 %
`04_floating_nut_carrier`	1	upside down: the 30 × 40 mm flange on the bed	40–60 %
`05_sample_platform`	1	largest flat face (not needed with the scan module)	30–50 %
`06_endstop_trigger`	1	broad pad, or turned so the horizontal M4 pilot hole prints cleanly	30–50 %

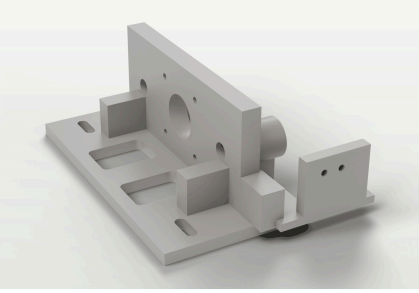


Figure 2. Motor-end block: rod bosses, coupler opening, switch tab.

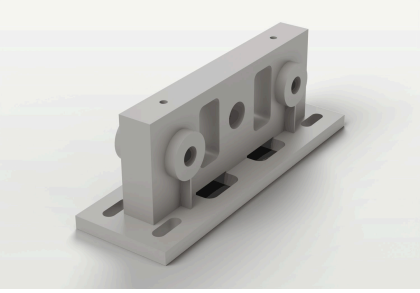


Figure 3. Far-end block.

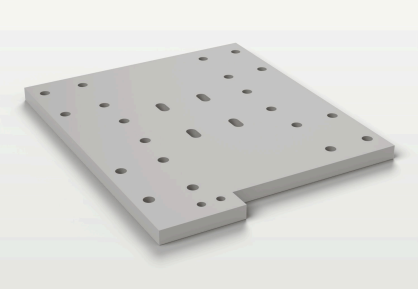


Figure 4. Carriage plate.

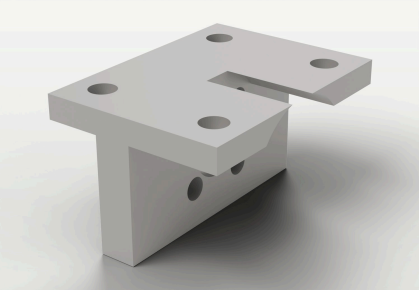


Figure 5. Floating nut carrier.

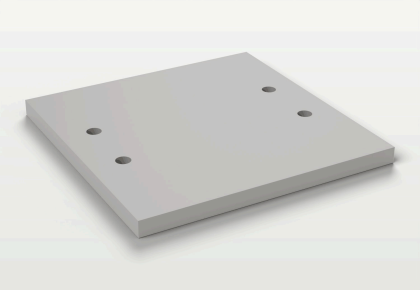


Figure 6. Sample platform.

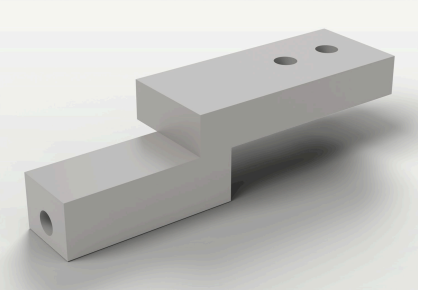


Figure 7. Endstop trigger.

2.3 Fasteners

Check every screw length against the printed parts before tightening.

Joint	Fasteners
Motor to motor-end block	4 × M3, 14–16 mm depending on washers; they must not reach the windings
Bearing blocks to carriage	up to 16 × M4 × 10–12
T8 nut to nut carrier	the nut's own 4 × M3 screws
Nut carrier to carriage	4 × M4 × 16, washers and nuts
Sample platform to carriage	4 × M4 × 16, washers and nuts
Trigger to carriage	2 × M3 × 16, washers and nuts
Trigger adjusting screw	1 × M4, about 20 mm, into the trigger's 3.3 mm pilot, with a nut to lock it
Rod retention	4 × M4, short, tapped or thread-formed into the 3.3 mm pilot holes
Home switch to the motor-end tab	2 small screws to suit the switch's holes (M2.5 or M3 class)

Joint	Fasteners
End blocks to the base	M5 bolts or wood screws with large washers, through the slots in the feet

2.4 Tools

Hex keys for M3, M4 and M5 and for the coupler's set screws; a small screwdriver; an M4 tap or a thread-forming screw for the pilot holes; a deburring tool or craft knife; pliers; a multimeter (continuity and resistance); a USB cable to the Pi; and a computer to prepare the Pi's SD card.

3 Printing

Setting	Value
Material	PLA (change material only once the fits are proven)
Nozzle, layer height	0.4 mm, 0.2 mm
Walls	at least 4
Top and bottom layers	at least 5
Infill	end blocks, carriage and nut carrier 40–60 %; platform and trigger 30–50 %
Supports	none, in the orientations of section 2.2

- Both end blocks have thin (0.8 mm) sacrificial fins under the two horizontal rod bosses, so the round bosses don't start in mid-air. Print them as part of the model and snap or trim them off afterwards.
- The openings in the walls are bridged. Don't add supports inside them unless a test print shows that your printer can't bridge them.
- Print a small part, or a slice of an end block, first and check the fits before printing both end blocks.

3.1 Preparing the parts

1. **Deburr.** Remove the elephant's foot, strings and any support residue from every mating face, and break the edges of the holes.
2. **Check the holes.** Try the M3, M4 and M5 clearance holes with their screws, without forcing them.
3. **Prepare the pilots.** Tap or thread-form the four 3.3 mm M4 rod-retention holes in the end blocks and the horizontal pilot in the trigger.
4. **Try the rods.** Push each rod through both end blocks. It should go in without force.
5. **Try the bearings.** Run every bearing block along both rods. Put aside any block that feels rough or has play.

4 Mechanical assembly

The rods guide, the screw pushes and the base supplies the stiffness. The nut carrier is only tightened once the carriage runs freely on its own; that order is what keeps the rail from binding.

4.1 Carriage and bearing blocks

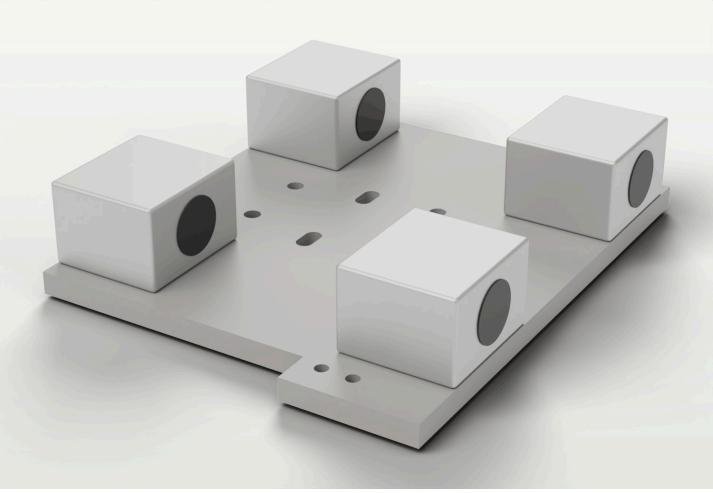


Figure 8. The carriage upside down with the four SC8UU blocks.

1. Lay the carriage upside down: the bearing face up, the platform face on the bench.
2. Place two SC8UU blocks on each rod line, 80 mm apart centre to centre.
3. Start all M4 screws by hand before tightening any of them.
4. Tighten them gradually and diagonally, so the blocks stay flat on the carriage. Leave them snug, not fully tight, until the carriage is on the rods.

The four bores must form two straight, parallel lines. A twisted block binds at once.

4.2 Rods and end blocks

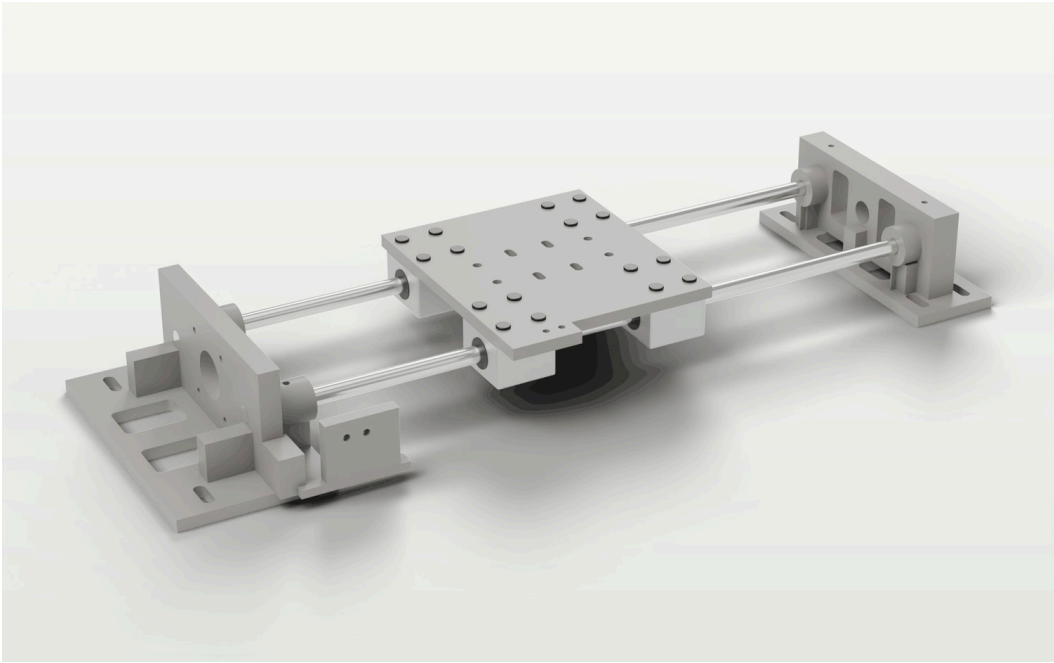


Figure 9. Both end blocks with the two rods and the carriage on them.

1. Stand the motor-end block on a known-flat surface (the base or a temporary one).
2. Push both rods through the motor-end block's bosses. Don't tighten the M4 retaining screws yet.
3. Support the carriage and slide it onto the rods through all four bearing blocks.
4. Slide the far-end block onto the rod ends.
5. Move the carriage back and forth and adjust the far-end block until the resistance is low and even over the whole travel. Then tighten the bearing-block screws fully.

6. Lock the rods: the motor-end retaining screws first, then the far end, gradually.

> **Check before going on**
> Without the lead screw, the carriage must slide smoothly over the whole travel. If it binds now, find the cause (table below) before fitting the screw.

Symptom	Likely cause	Correction
Tight everywhere	rod spacing does not match the blocks	loosen the far block and the bearing screws, let them settle, tighten again
Tight near one end	rods not parallel	realign the far block
Rough at regular intervals	dirty or damaged rod or bearing	clean, turn the rod, or replace the block
Carriage rocks	loose block or worn bearing	tighten the block, or replace it

4.3 Motor and coupler

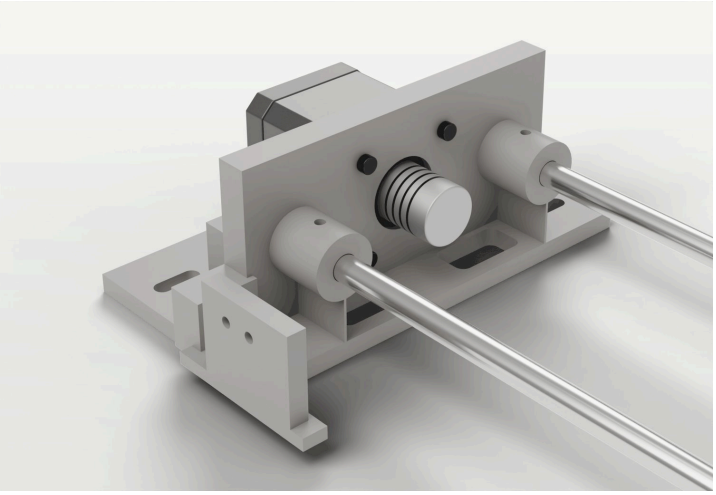


Figure 10. The NEMA 17 on the motor-end block, with the coupler in the opening.

1. Fit the coupler to the motor shaft and tighten its motor-side set screw on the shaft's flat.
2. Bolt the motor to the outer face of the motor-end plate with four M3 screws, the coupler passing through the central opening. Check that the screws don't bottom out in the motor.
3. Turn the motor by hand until both set screws can be reached through the opening.

4.4 Lead screw and nut carrier

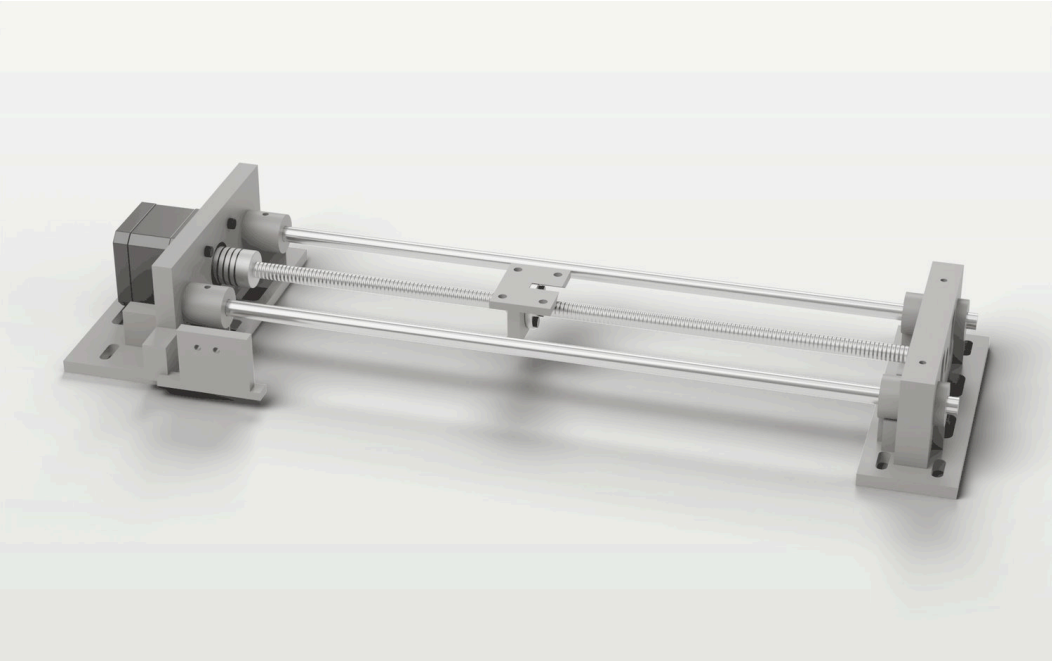


Figure 11. The lead screw in the coupler and the T8 nut on the floating carrier (carriage hidden).

1. Screw the T8 nut's flange to the vertical face of the floating nut carrier with its four M3 screws.
2. Thread the nut onto the lead screw by turning the screw, never forcing the thread.
3. Push the screw about 6–7 mm into the coupler, without letting it touch the motor shaft, and tighten the screw-side set screw. The far end of the screw stays free, clear of the far-end block.
4. Bolt the carrier under the carriage with four M4 × 16 through the adjustment slots, loosely.
5. Run the carriage through its whole travel by hand, turning the screw, so the carrier finds the screw's natural line.
6. Tighten the carrier evenly at the position where the screw turns with the least side load, without pulling the carriage sideways.

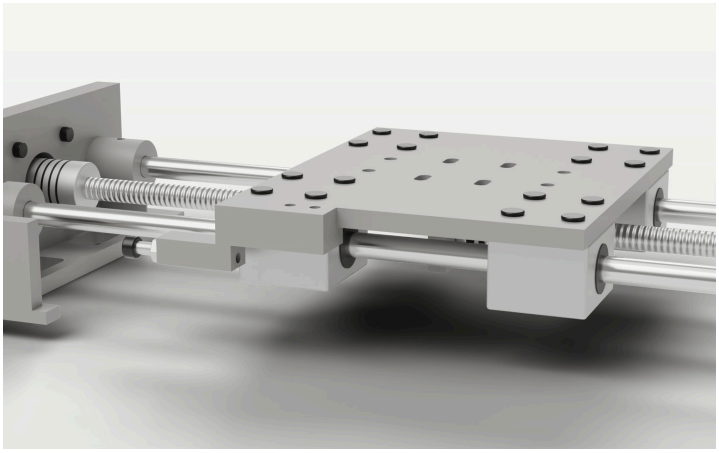


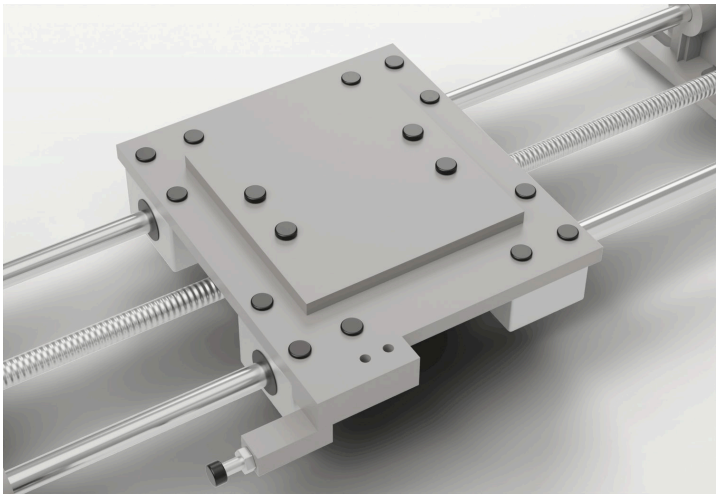
Figure 12. The carriage over the nut carrier, seen from low down (platform not fitted).

The screw is not self-locking

A Tr8×8 screw moves 8 mm per turn and can be back-driven: with the motor off, the carriage can be pushed. Turning the motor off therefore clears the rail's homed state in the app.

Check now: the screw turns by hand over the full travel, the carriage doesn't move sideways as it turns, and the coupler runs true.

4.5 Sample platform



Bolt the platform to the carriage's 60 × 20 mm hole pattern with four M4 × 16, washers and nuts. Tighten until it is rigid, without bowing either plate.

If you are fitting the scan module, leave the platform off: the module bolts to the same four holes.

Figure 13. The sample platform on the carriage.

4.6 Home switch and trigger

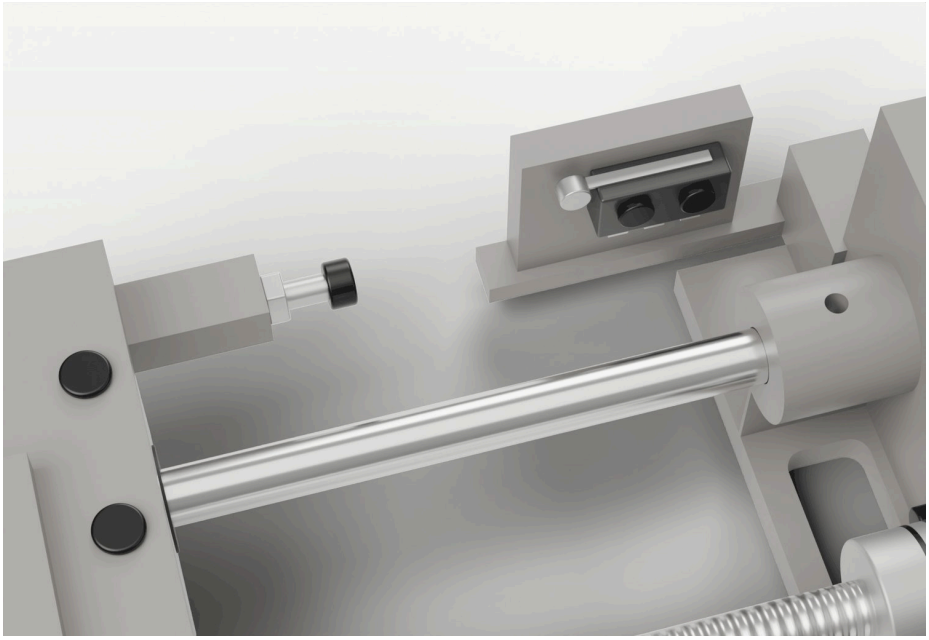


Figure 14. The roller switch on the motor-end tab and the trigger with its adjusting screw. In the renders the switch's position is approximate; follow the steps.

Note that a roller lever opens again well past the point where it closed: 3.8 mm on the prototype. The firmware allows for that (section 9), so this is normal.

1. Screw the roller microswitch to the tab on the motor-end block, its roller facing the carriage, so the trigger approaches the roller gradually.
2. Bolt the trigger to the carriage with two M3 × 16, washers and nuts.
3. Turn the M4 adjusting screw into the trigger's horizontal pilot, with its lock nut.
4. Move the carriage slowly towards the motor end by hand and listen for the switch's click.
5. Set the adjusting screw so the switch clicks before any part of the carriage, the bearing blocks or the nut carrier reaches the motor-end block, and the lever still has travel left after the click.
6. Lock the screw with its nut once the switch triggers reliably.

4.7 Mounting on the base

1. Fix the motor-end block to the base with M5 bolts or screws and large washers.
2. Leave the far end loose and run the carriage end to end a few times so the rods settle.
3. Tighten the far end last, alternating between the screws, and check that the motion has not become tighter.

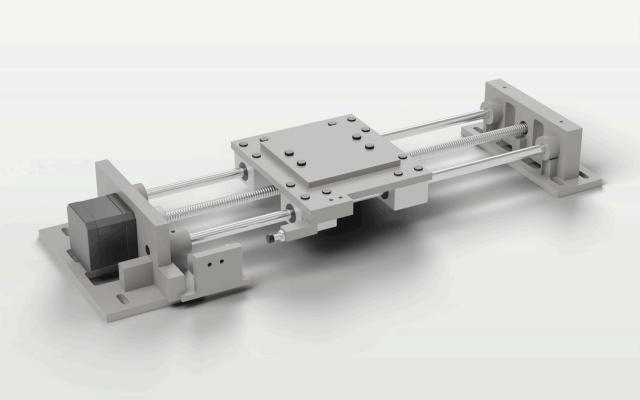


Figure 15. The finished rail from the motor end.

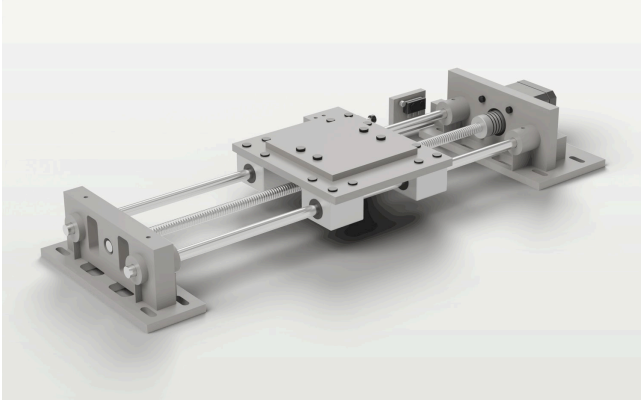


Figure 16. The finished rail from the far end.

4.8 Mechanical checks

- [ ] The rods are clean, straight and held firmly.
- [ ] All four bearing blocks sit flat and are tight; none feels rough or binds.
- [ ] The lead screw turns without steering the carriage sideways, and the coupler set screws are tight and reachable.
- [ ] The nut carrier is tight without side-loading the screw.
- [ ] The switch clicks before any hard contact, with overtravel left.
- [ ] At least 15 mm of clearance remains at the far soft limit (140 mm from the switch).
- [ ] The platform is rigid and doesn't rock; the end blocks are firmly mounted on a heavy, flat base.

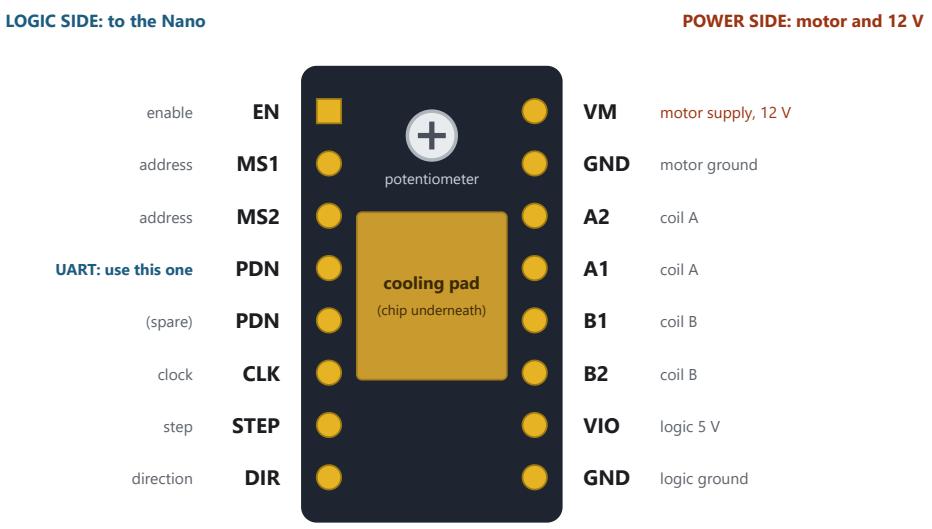
5 Electronics

The controller is built on a half-size breadboard: an Arduino Nano, a TMC2209 driver module, two resistors and a capacitor. This section introduces the parts; section 6 puts them on the board.

Connections. Pins are constants in `firmware/src/config.h`.

Nano	Goes to	Notes
D2	home switch NO (switch C to GND)	closes (LOW) when pressed; hardware interrupt INT0
D3	TMC2209 STEP	one pulse is one 1/16 microstep, 2.5 µm
D4	TMC2209 DIR	
D5	TMC2209 EN, plus 10 kΩ to 5 V	low enables the motor; the pull-up keeps it off while the Nano starts
D7	TMC2209 PDN_UART, direct	UART receive
D8	TMC2209 PDN_UART, through 1 kΩ	UART transmit; a single-wire UART needs the resistor
5V, GND	TMC2209 VIO, GND	logic supply; ground common with the 12 V supply
–	TMC2209 MS1, MS2 to GND	UART address 0
–	12 V to VM, 12 V − to GND	motor supply, with 100 µF across VM and GND at the module
D9	nothing	reserved for a shutter-cable relay

5.1 TMC2209 driver

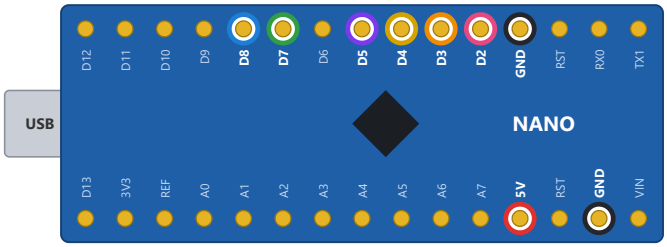


Top view: potentiometer up, pins down into the breadboard. Check the labels on your board.

Figure 17. The usual TMC2209 module pinout (BTT V1.2/V1.3 and most clones).

- [ ] Turn the module over and compare the labels next to its pins with the drawing. Names vary: *OA1/OA2* or *1A/1B* for A1/A2, *VDD* or *VCC* for VIO.
- [ ] Find the UART pin: labelled PDN, PDN_UART or UART. On some BTT modules a solder jumper must be bridged to bring it out to the header.
- [ ] Check the sense resistors (marked R100 or R110: 0.11 Ω). If yours differ, change `DRIVER_RSENSE_OHM` in `firmware/src/config.h`: the current scales with it.
- [ ] The potentiometer (VREF) is not used: the current is set over UART from the app.
- [ ] If the module came with a heatsink, stick it on the pad on top, clear of all pins.

5.2 Arduino Nano



D2	home switch (interrupt INT0)
D3	STEP
D4	DIR
D5	EN, active low
D7	UART receive (direct)
D8	UART transmit (through 1 kΩ)
5V	logic supply
GND	ground (both pins)

Figure 18. The Nano as it sits on the breadboard, USB to the left. The ringed pins are used.

The rail uses six signal pins (D2–D5, D7, D8), 5V and both GND pins. D2 is the hardware-interrupt pin, so the switch stops the motor before its next step. A CH340 clone is fine; the Pi's USB powers it. The same colours are used for each signal in every drawing and table.

5.3 Resistors, capacitor and switch



Figure 19. R2, 1 kΩ: brown, black, red (5-band: brown, black, black, brown). D8 transmits through it.



Figure 20. R1, 10 kΩ: brown, black, orange (5-band: brown, black, black, red). Pull-up on EN.

Resistors have no direction. If unsure of the colours, measure them: about 1.0 kΩ and 10 kΩ.

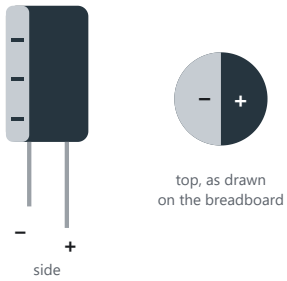


Figure 21. C1, 100 µF.

C1 sits across VM and GND next to the driver and absorbs voltage spikes from the motor. The stripe marks minus (−); the longer leg is plus (+). It must go in the right way round: − to GND, + to VM. Use a 25 V or higher rating.

The home switch has three terminals: C (common), NO (normally open) and NC (normally closed). Wire C and NO, so the switch closes when pressed and pulls D2 to ground; NC stays empty.

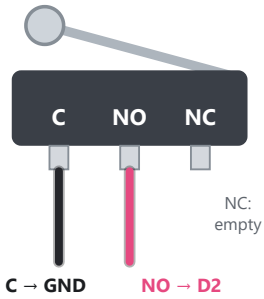


Figure 22. The home switch, wired normally open.

The switch is armed during every move, not only while homing. Its first contact stops any motion before the next step. While it is pressed, moves towards it are refused.

To wire it normally closed (C and NC) instead, set `ENDSTOP_ACTIVE_LEVEL` to `HIGH` in `firmware/src/config.h` and reflash. A broken wire then also reads as pressed, which fails safe, at the cost of a stop whenever a connector works loose.

5.4 Motor coils

The NEMA 17 has four wires: two coils. With a multimeter on Ω, the two wires of one coil read a few ohms; wires of different coils read open. One pair goes to A1/A2, the other to B1/B2. Wire colours vary between motors, so measure, and mark the pairs before wiring.

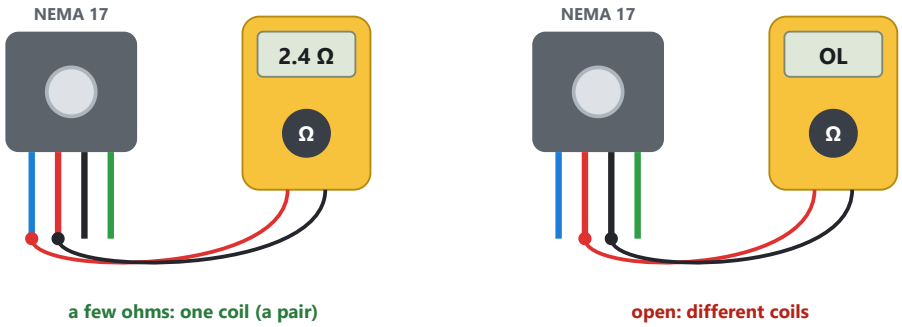


Figure 23. Finding the coil pairs: touch two wires at a time until both pairs are found.

6 Wiring on the breadboard

6.1 The breadboard

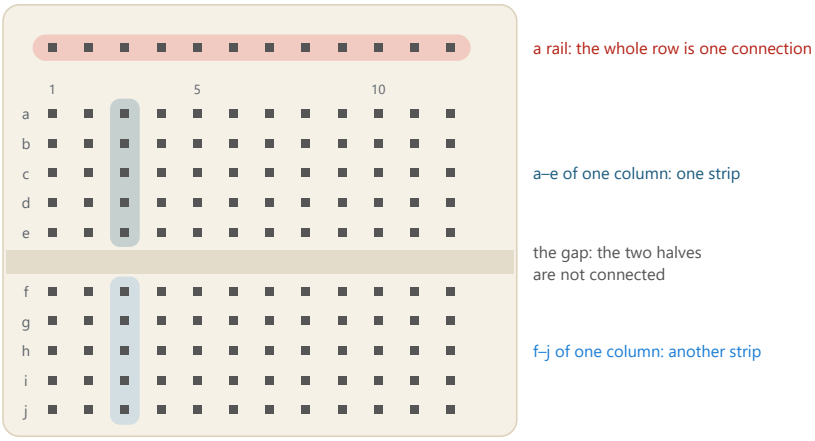


Figure 24. Which holes are joined inside a breadboard.

- Columns are numbered 1–30 and rows lettered a–j. A hole is named by row and column, as in `c13`.
- Two parts are connected when their legs are in the same five-hole strip. Modules straddle the middle gap so that each pin has its own strip.
- Follow your board's red (+) and blue (−) markings for the rails; their order varies between makers. Long boards sometimes split the rails in the middle; this build stays within columns 1–30.

6.2 Reading the wiring drawing

- Each wire carries a label such as **W12 D7 → PDN**: its number in the wire table, where it starts and where it ends.
- Most signal wires arch over the board so they don't cross. W9 and W10 lie flat in rows c and b, and the switch lead W14 runs flat between rows a and b.
- Wire colours don't matter electrically, but matching the drawing makes checking easier: black for ground, red for 5 V, one colour per signal.
- The 12 V leads go straight to the driver's VM and GND pins and never touch the rails, which carry the Nano's 5 V.

> **Before you start**
> USB unplugged, 12 V supply off and disconnected, motor not connected.

6.3 Complete wiring

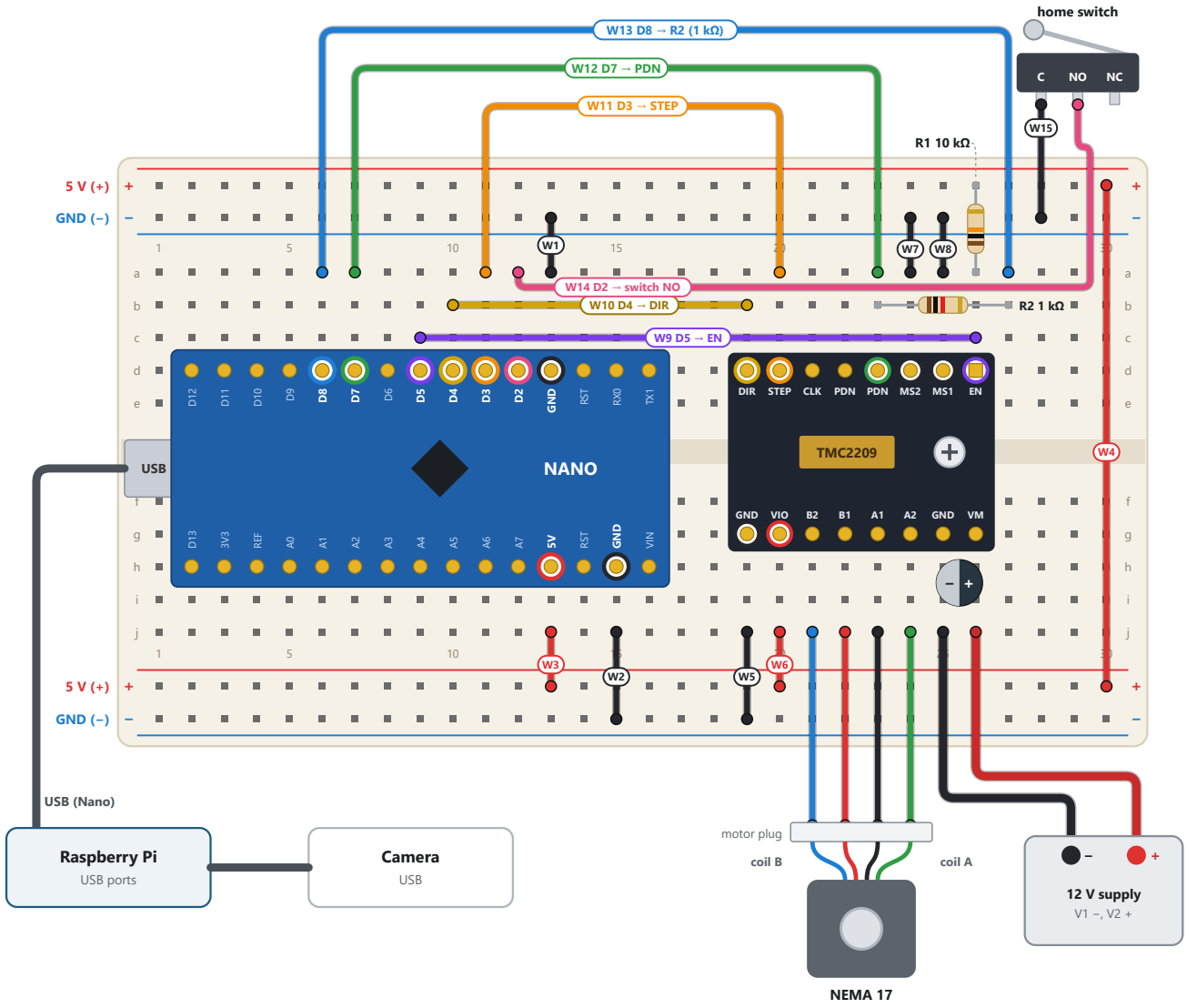


Figure 25. The complete controller. The TMC2209 is the module of section 5.1, turned a quarter turn clockwise: logic pins along the top, power pins along the bottom. The motor and the 12 V leads go on last, with the supply off (section 9).

6.4 Wire table

Work down the table one line at a time and tick each line off against the drawing. The top rails are above row a, the bottom rails below row j.

	No.	From	To	Note
1. Boards				
☐	**P1**	Arduino Nano: pins in rows d and h, columns 2–16		D12 and D13 in column 2, USB to the left
☐	**P2**	TMC2209: pins in rows d and g, columns 19–26		DIR in d19, EN in d26, VM in g26; potentiometer to the right
2. Wires and parts				
☐	**W1**	a13 (Nano GND, top row)	top GND rail	
☐	**W2**	j15 (Nano GND, bottom row)	bottom GND rail	
☐	**W3**	j13 (Nano 5V)	bottom 5 V rail	
☐	**W4**	bottom 5 V rail	top 5 V rail	column 30, joins the rails
☐	**W5**	j19 (TMC GND)	bottom GND rail	logic ground
☐	**W6**	j20 (TMC VIO)	bottom 5 V rail	logic supply
☐	**W7**	a24 (TMC MS2)	top GND rail	UART address 0
☐	**W8**	a25 (TMC MS1)	top GND rail	UART address 0
☐	**R1**	a26 (TMC EN)	top 5 V rail	10 kΩ pull-up
☐	**W9**	c9 (D5)	c26 (EN)	flat on the board
☐	**W10**	b10 (D4)	b19 (DIR)	flat on the board
☐	**W11**	a11 (D3)	a20 (STEP)	
☐	**W12**	a7 (D7)	a23 (PDN, UART)	UART receive, direct
☐	**R2**	b23 (PDN)	b27	1 kΩ
☐	**W13**	a6 (D8)	a27	UART transmit, through R2
☐	**W14**	switch NO	a12 (D2)	between rows a and b
☐	**W15**	switch C	top GND rail	NC stays empty
☐	**C1**	100 µF: stripe (−) in h25	+ in h26	check the polarity
☐	**U1**	Nano USB	Raspberry Pi	a data cable
☐	**U2**	Camera USB	Raspberry Pi	
3. Last, with the 12 V supply off (section 9)				
☐	**M1**	motor coil B	j21 (B2), j22 (B1)	
☐	**M2**	motor coil A	j23 (A1), j24 (A2)	
☐	**V1**	12 V supply −	j25 (TMC GND)	straight to the pin
☐	**V2**	12 V supply +	j26 (TMC VM)	straight to the pin

6.5 Checks before power

- [ ] **Coil pairs:** a few ohms within each pair, open between the pairs; one pair on A1/A2, the other on B1/B2.
- [ ] **UART:** W12 and R2 land on the UART pin (PDN, PDN_UART or UART), with the solder jumper bridged if your module needs it.
- [ ] **Switch:** C to the GND rail, NO to D2, NC empty.
- [ ] **Common ground:** with the meter on continuity, j25 (driver motor ground) and the bottom GND rail beep: the module joins its two GND pins.
- [ ] **No short:** the 5 V rails and the GND rails don't beep.
- [ ] **C1:** stripe (−) in h25, + in h26.

7 Firmware

The Nano's firmware is in `firmware/` and is built with PlatformIO. It can be flashed from any computer with PlatformIO installed, or on the Pi once the app is installed (section 8).

7.1 From a computer

```
pio run -d firmware -e nano -t upload
```

Use `-e nano_new` instead for a genuine Nano or one with the new bootloader; most CH340 clones use the old one (`nano`).

7.2 On the Pi

The app owns the serial port, so stop it first:

```
uv tool install platformio
sudo systemctl stop macrorail
~/.local/bin/pio run -d ~/macrorail/firmware -e nano -t upload --upload-port /dev/ttyUSB0
sudo systemctl start macrorail
```

The first build downloads the compiler and libraries, which takes several minutes on a Pi.

7.3 Settings in the firmware

Everything that depends on the wiring is a constant in `firmware/src/config.h`:

Constant	Default	Change when
`INVERT_DIRECTION`	`true`	the carriage moves towards the switch on *Away*
`ENDSTOP_ACTIVE_LEVEL`	`LOW`	the switch is wired normally closed (then `HIGH`)
`DRIVER_RSENSE_OHM`	`0.11`	the driver module has other sense resistors

8 Raspberry Pi and the app

8.1 Preparing the Pi

1. With Raspberry Pi Imager, write *Raspberry Pi OS Lite (64-bit)* to the SD card. In its settings, set a host name (for example `macrorail`), a user and password, the Wi-Fi network, and turn on SSH.
2. Start the Pi from its 2.5 A supply and log in over SSH: `ssh <user>@macrorail.local`.
3. Install git: `sudo apt update && sudo apt install -y git`.

The Lite image is preferred: desktop images run a gphoto2 volume monitor that can grab the camera. The installer warns if it finds one.

8.2 Installing the app

```
git clone https://github.com/Sam-de-Ham/MacroRail.git ~/macrorail
~/macrorail/deploy/install.sh
```

The script installs uv and the Python dependencies, creates `~/.config/macrorail/config.toml` from `deploy/config.example.toml`, and installs and starts the `macrorail` service, which then also starts on every boot. It asks for your password for the `sudo` steps, and warns if your user is not in the `dialout` and `plugdev` groups (needed for the serial port and the camera). Running it again later updates an existing installation and keeps the config.

Open **http://macrorail.local:8000** (your host name) on a phone or laptop on the same network.

Task	Command
Follow the log	`journalctl -u macrorail -f`
Restart after a config change	`sudo systemctl restart macrorail`
Turn autostart off or on	`sudo systemctl disable --now macrorail` (`enable --now`)
Install without autostart	`~/macrorail/deploy/install.sh --no-autostart`
Update	`cd ~/macrorail && git pull && deploy/install.sh`

8.3 The config file

`~/.config/macrorail/config.toml` holds the hardware constants; day-to-day settings (speeds, currents, step size) are changed in the app. The defaults suit the rail as built:

```
[rail]
port = "auto"            # finds the Nano by what it reports
steps_per_mm = 400       # 200 steps x 16 microsteps / 8 mm lead
min_mm = 0.5             # soft limits, mm from the switch
travel_mm = 140.0
homing_speed_mm_s = 2.0
```

Leave `port` on `"auto"`. It also works when the scan module's Nano is plugged in, which a `/dev/serial/by-id/...` path does not (both Nanos share that name). Restart the app after any change.

> **Never pull the plug on a running app**
> Stop or restart it with `systemctl`. An unclean exit can leave the A6400 unresponsive until it is switched off and on again.

9 First power-up and homing

Do these checks in order, the first time and after any change to the wiring. Keep a hand near the 12 V switch.

1. **Driver, without the motor.** Motor unplugged, 12 V on, the Nano on the Pi. The rail's status in the app's top bar shows *Not homed*. *No driver power* means the TMC2209 doesn't answer: check 12 V, the UART wiring (W12, W13, R2) and MS1/MS2 (W7, W8). The driver only answers on UART while VM has power.
2. **Switch.** Press the switch by hand. The Position card on the Shoot tab shows *Home switch pressed* at once.
3. **Motor.** 12 V off, plug in the motor (M1, M2), 12 V on.
4. **Direction.** On the Shoot tab, choose 1 mm and press *Away*. The carriage must move away from the switch. If it moves towards it, swap the two wires of one coil (12 V off), or change `INVERT_DIRECTION` and reflash.
5. **Home.** Press *Home*. The rail moves towards the switch at 2 mm/s and stops on its first contact, backs off until the switch has opened again plus 1 mm, and touches it again at the same speed: that contact is position 0. It then parks 0.5 mm from the switch, or just clear of the release point if that is further. The activity log reports the release distance.
6. **Travel.** Jog to the far end of the travel and check that the carriage stops at the soft limit with at least 15 mm to spare. If your rail is shorter, lower `travel_mm`.

9.1 Motor current and backlash

The motor current is set on the System tab under Motion → Motor current: 600 mA running and 200 mA holding by default. The full current is applied from the first move until the rail has been idle for 30 s, and is never changed during a stack. Raise it only if the motor misses steps, and check that the motor stays comfortably warm.

Once per setup, with the camera running (section 10) and a small, sharp detail in focus, press **System → Measure backlash**. The rail passes through focus in both directions, measures the play in the screw and offers a take-up distance (the default is 0.3 mm until measured; the prototype measured 14 µm and uses 0.1 mm). Every arrival at a stack's start then ends with a move in the stack direction.

10 Camera

The app controls the camera over USB with gphoto2. It was developed with a Sony A6400; other bodies that gphoto2 can trigger remotely should work, with their own menu names.

10.1 Sony A6400 settings

Menu item	Setting	Why
USB Connection	PC Remote	remote capture over USB
PC Remote Settings → Still Img. Save Dest.	PC+Camera	the full-size files stay on the card; a copy comes over USB
PC Remote Settings → PC Save Img	JPEG Only	a small JPEG for the app's previews, also when shooting RAW
USB Power Supply	Off	charging over USB overloads the Pi's supply and drops USB devices
Auto Power OFF	off, or the longest time	long stacks
Focus mode, exposure	manual focus, manual exposure	the rail does the focusing; every frame exposed alike
Shutter type	electronic (silent) where possible	avoids shutter shock at high magnification

Connect the camera to the Pi; its status in the top bar turns green. Live view appears on the Shoot tab. Use a dummy battery for long stacks: the battery drains while the camera is connected.

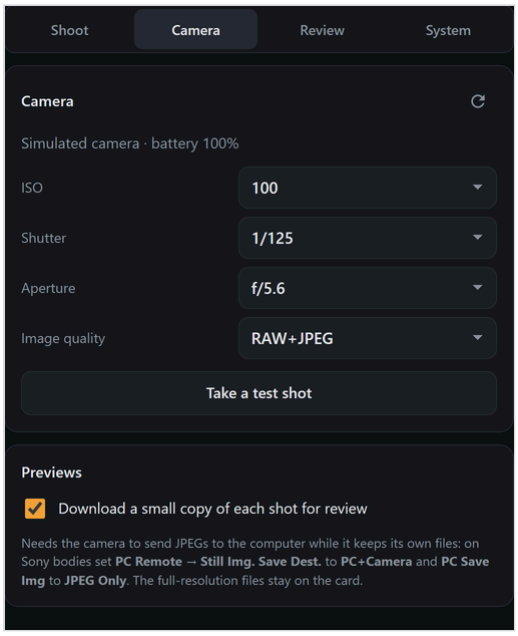


Figure 26. The Camera tab: ISO, shutter, aperture, exposure compensation, white balance, quality, size and focus mode can be set from the browser.

The camera's settings can be changed on the Camera tab. The previews option downloads a small copy of each shot for review on the Pi; it needs the two PC Remote settings above.

A capture is confirmed by the camera in about 1.4 s on the A6400. If a shot fails, the app reconnects and retries once; a second failure pauses the stack, and Resume retakes that slice. A duplicate frame is possible, a missing one is not.

11 Shooting a stack

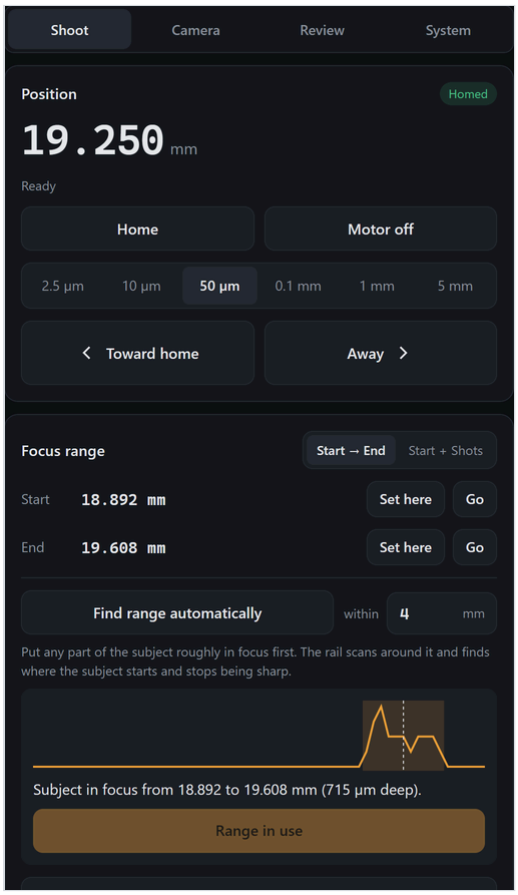


Figure 27. The Shoot tab: position, jog, focus range and stack settings.

1. **Home** the rail.
2. **Set the range.** Jog to the front of the subject on the live view and press *Set here* for Start, then the same for End. Or bring any part of it roughly into focus and press *Find range automatically*: a focus scan finds where the subject starts and ends. Tap the live view to zoom in 3×.
3. **Choose a step.** *Which step?* suggests half the depth of field from the magnification and numerical aperture (for a 4× / 0.10 objective, 32.5 µm). The Start button shows the number of shots.
4. **Preview range** moves between the two ends so you can check the framing.
5. **Start stack**, optionally with a name. Pause holds the position; Stop returns to the start.
6. **Review** shows each stack's shots, a quick all-in-focus preview and a rough depth map. *Download* gives a zip of the record, the preview, the depth maps and the small copies.

One device controls the rail at a time; another can take over with *Take control*. Stop works from any device. Full-size files stay on the camera's card: stack them with your stacking software (Zerene Stacker, Helicon Focus or similar).

12 Troubleshooting

Symptom	Likely cause and fix
Mechanics	
Carriage binds with the screw removed	Rods not parallel or spacing wrong: loosen the far block and the bearing screws, realign (section 4.2).
Binds only with the screw fitted	Nut carrier tightened off the screw's line: loosen it, run the carriage end to end, tighten again.
Tight spot once per turn of the screw	Bent screw or coupler not centred: check the screw's insertion and the coupler; replace a bent screw.
Uneven slices in a stack	Backlash not taken up: run *Measure backlash* and apply its take-up; check the nut carrier and coupler set screws.
Controller	
Status shows *No driver power*	12 V missing at VM, or the UART is miswired: check W12, W13, R2 and MS1/MS2. The TMC2209 answers only while VM is powered.
The rail is not found	`port = "auto"` in the config, the firmware flashed, a USB data cable (not charge-only). `ls /dev/serial/by-path/` must list it.
Moves towards the switch on *Away*	Swap the two wires of one coil (12 V off), or change `INVERT_DIRECTION` and reflash.
Motor hums or vibrates without turning	Coil pairs mixed up: measure them again (section 5.4).
Homing fails with `NO_SWITCH`	The switch never closed: check C to GND and NO to D2, press it by hand (the Position card must show it), and check that the trigger reaches the roller.
Homing fails with `SWITCH_STUCK`	The switch stayed closed after backing off 5 mm: wired C and NC, or the trigger screw set too far out.
A move stops with `ENDSTOP`	The switch closed during a move: the trigger reaches the roller too early, or a loose switch wire picks up noise.
`LINK` errors, motion stops	No message from the Pi for 2 s: loose USB cable, or the Pi under-powered. Use a 2.5 A supply.
Camera and Pi	
Camera not found	USB Connection not on PC Remote, or a desktop image's gphoto2 monitor holds the camera. Unplug and replug the camera.
Camera drops out during a stack	The Pi is under-voltage (the System tab and the log warn): set the camera's USB Power Supply to Off and use a 2.5 A supply.
Camera stops responding after a crash	Switch the camera off and on. Always stop the app with `systemctl`.
No previews in Review	Set Still Img. Save Dest. to PC+Camera and PC Save Img to JPEG Only.
Can't install the app on a phone, no system notifications	Browsers only allow these over HTTPS or on the Pi itself. Over plain HTTP the stack alerts still chime, vibrate and mark the tab title.

Further reference: `docs/wiring.md` (pin map), `docs/protocol.md` (the serial protocol and its error codes) and `docs/design-notes.md` (why things are done the way they are).